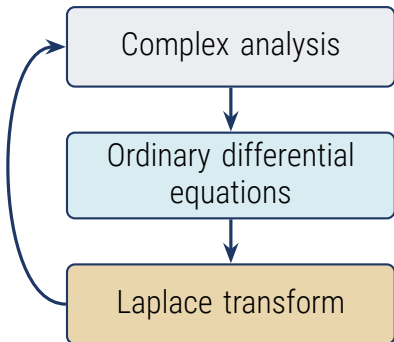




MAT290: Advanced Engineering Mathematics

Lecture 1: An Introduction

Erfan Meskar

University of Toronto – LEC0101

Fall 2026



UNIVERSITY OF
TORONTO

Welcome to Second Year!

- Erfan Meskar — LEC0101

- e.meskar@utoronto.ca

- Department of Electrical and Computer Engineering, University of Toronto

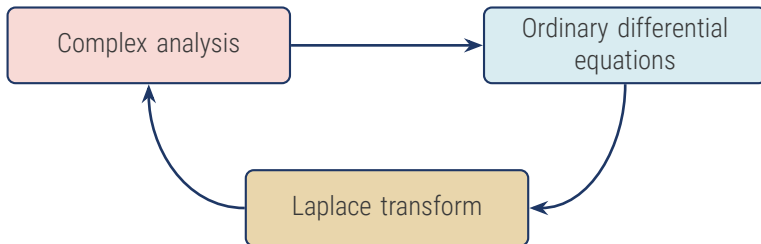
- **Research:** game theory, machine learning, and computer networks

Other instructors

- Prof. Manfredi Maggiore — SEC0102

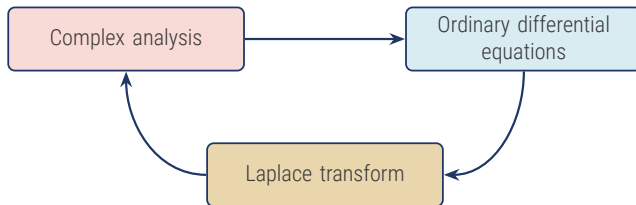
- Prof. Adrian Nachman — SEC0103 and SEC0104

What This Course Is About

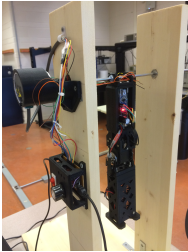


- You've seen complex numbers and some ordinary differential equations in **ECE110 (Electrical Fundamentals)**.
- You'll use them this year in **ECE231 (Introductory Electronics)** and **ECE221 (Electric and Magnetic Fields)**.

- You've seen complex numbers and some ordinary differential equations in **ECE110 (Electrical Fundamentals)**.
- You'll use them this year in **ECE231 (Introductory Electronics)** and **ECE221 (Electric and Magnetic Fields)**.
- There is a whole world of engineering that requires these three areas:



- This course is the **door to that world**.



A robot gymnast (UofT)



A snake robot (NTNU Trondheim)

- These robots use **servomotors**.
- **Servomotor**: a motor with an angle sensor and an embedded algorithm to adjust the shaft angle.

Examples shared by Prof. Manfredi Maggiore.



Servomotor

Mathematical model

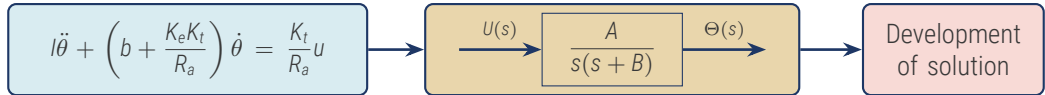
$$I\ddot{\theta} + \left(b + \frac{K_e K_t}{R_a}\right) \dot{\theta} = \frac{K_t}{R_a} u$$

- θ is the shaft angle.
- u is the motor voltage.

Problem

Decide what $u(\theta(t))$ should be to make $\theta(t) \rightarrow \theta_{\text{des}}$.

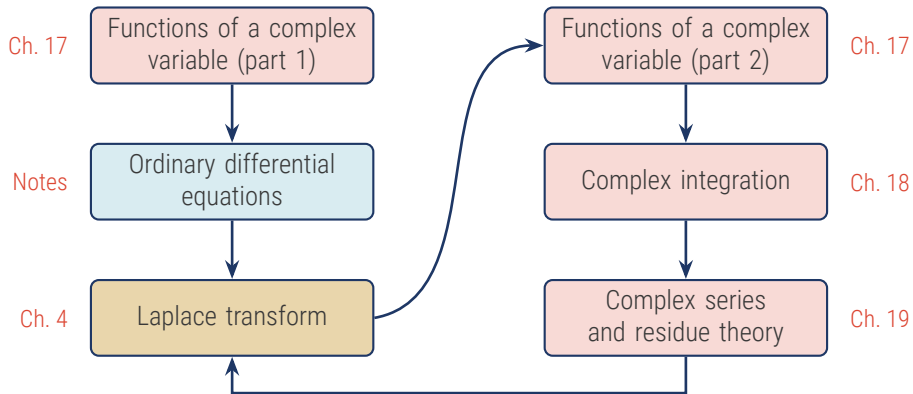
Solution technique



- Controlling one motor is easy.
- How do we coordinate 20 servos to make a humanoid robot walk?
- This is a hard problem.
- You need the mathematical tools you will learn this semester in **MAT290 and MAT291!**



Our Progression in This Course



■ Five quizzes, each worth 10%

- **Dates:** October 1, October 15, November 5, November 19, and December 3
- Held during the **Thursday tutorials, 2-3 PM**. You **MUST** go to your registered Thursday tutorial room.
- Duration: 45 minutes
- One question is taken from the assigned homework problems
- Aid-sheet **NOT** allowed. Non-programmable calculator allowed.
- Regrade request: see the Syllabus
- Missed quizzed with an approved petition: see the Syllabus

■ Final examination worth 50%

Minimum Final Exam Requirement

Students must achieve a minimum grade of **40% on the final examination**, regardless of their performance on other course components. Students who do not achieve at least 40% on the final exam will not pass the course, and a grade of **49%** will be assigned, regardless of their overall course average.

■ Homework Assignments:

- Problem sets will be assigned each week
- They are designed to build fluency with the course material and prepare you for the quizzes and final exam.
- Homework is **not submitted or graded**.

■ Tutorials:

- Tutorials begin in **Week 3**, with two one-hour sessions each week.
- Tutorials focus on problem solving, discussion of assigned problems, and quiz preparation.
- On weeks without a scheduled quiz, Thursday tutorial session will run as regular tutorials.

How to Get The Most Out of This Course

1. Attend lectures and take notes actively.
2. Review the assigned textbook/lecture-note readings after each lecture.
3. Attempt the assigned problems **on your own before the tutorials**.
4. Attend tutorials, compare your approach with the solutions, and **ask questions**.

Office hours

- Very important! We will schedule them today
- Times and location will be announced on Quercus.

Communication

- Piazza is not used in this course.
- We value personal engagement.
- Communicate through office hours and tutorials.
- Don't hesitate to email me at e.meskar@utoronto.ca if you have an issue.

- **Required textbook:** Dennis G. Zill, *Advanced Engineering Mathematics*, 7th edition
- The course follows it closely. Homework uses the **7th edition** (the 6th edition won't work).
- For Unit 2 (ODEs), we will use the **posted course notes**.

Lectures	Topic covered	Readings	Homework assignments
1–3	Unit 1: Complex numbers Complex numbers; complex exponential and logarithm	17.1, 17.2, 17.6	Section 17.1: 5, 27, 33, 42, 47, 48. Section 17.2: 3, 18, 21, 30, 34, 39. Section 17.6: 11, 17, 24, 32, 38, 47.
4–6	Complex trigonometric and hyperbolic functions; Unit 2: ODEs Introduction to ordinary differential equations	17.7 ODE notes: Sec. 1.1–1.3	Section 17.7: 2, 12, 15, 21, 23, 31. Review (p. 862): 3, 10, 15.
7–9	Existence and uniqueness, equilibria and stability	ODE notes: Sec. 1.3–1.5, 2.1	ODE notes: Problems I.1–I.15 in Sec. 1.6

Have a Great Semester!